A Comparative Study of Autoencoder-Based Generative Models for One-Class Anomaly Detection

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Abstract

We study unsupervised anomaly detection, where a model is trained only on normal data and must flag anomalous samples at test time. Rather than contrasting unrelated model families, we investigate a single, coherent question: *how does the way a model regularizes its latent space affect anomaly detection?* We compare three autoencoder-based generative models that form a spectrum of increasingly explicit latent-space regularization: a plain convolutional autoencoder (AE, no latent constraint), a variational autoencoder (VAE, a KL term pulling the latent toward a Gaussian prior), and an adversarial autoencoder (AAE, an adversarial game matching the latent to that prior). All three share the same encoder–decoder backbone, are scored by reconstruction error, and are evaluated under one protocol on a one-class reformulation of Fashion-MNIST and on the MVTec AD industrial-defect benchmark, using AUROC and PR-AUC. Our AE baseline attains a mean AUROC of 0.90 across the ten Fashion-MNIST classes, establishing a strong reference against which the VAE and AAE are assessed. For the VAE, we add an explicit KL-regularized latent model and evaluate both deterministic reconstruction-error scoring and Monte-Carlo reconstruction-probability scoring under a β-ablation. The preliminary class-0 VAE run included in the repository reports AUROC 0.897 and PR-AUC 0.933 with reconstruction-error scoring; these headline numbers should be replaced by the final `scripts/run_vae_ablation.py` output once the full run is completed. The AAE integrates adversarial latent regularization into the same reconstruction-based pipeline and remains competitive, reaching AUROC 0.897 on the shared setup. Across the three models the differences in AUROC are within roughly one percentage point, indicating that latent-space regularization is approximately neutral for reconstruction-based scoring on this benchmark.

1 Introduction

Anomaly detection is the task of identifying data points that deviate from a notion of "normal." It matters wherever anomalies are rare, costly, and hard to enumerate in advance — industrial quality control, medical screening, fraud and intrusion detection. The defining difficulty is asymmetry: normal data is abundant, whereas anomalies are scarce, diverse, and often unknown at training time. This makes ordinary supervised classification ill-suited and motivates a *one-class* approach: learn the structure of normal data alone, then treat poorly-fitting samples as anomalous.

Autoencoders are a natural fit: trained to compress and reconstruct normal data, they reconstruct unfamiliar inputs poorly, and the reconstruction error becomes an anomaly score. But a plain autoencoder places no constraint on its latent space, which can let it generalize too well and reconstruct anomalies it should reject. This motivates *regularizing the latent representation.* In this work we ask a focused question: *does shaping the latent space toward a known prior improve* *reconstruction-based anomaly detection, and does the way we enforce that prior matter?* We study three autoencoder-based models that answer "how to enforce the prior" differently, forming a clean progression:

$$\underbrace{\text{AE}}_{\text{no latent constraint}} \longrightarrow \underbrace{\text{VAE}}_{\text{KL to a Gaussian prior}} \longrightarrow \underbrace{\text{AAE}}_{\text{adversarial match to a prior}}.$$

By fixing the dataset, the preprocessing, the encoder–decoder backbone, the scoring convention, and the metrics, we isolate the single variable of interest: the latent-regularization strategy.

Our contributions are: (i) a unified, reproducible evaluation framework in which any one-class autoencoder is scored identically; (ii) a strong AE baseline characterized across all Fashion-MNIST classes; and (iii) a controlled comparison of AE, VAE, and AAE on Fashion-MNIST and MVTec AD that reads latent regularization as the explanatory factor.

2 Related Work

Reconstruction-based anomaly detection with autoencoders is long established: a network trained to compress and reconstruct normal data fails to reconstruct unfamiliar inputs, and the reconstruction error serves as an anomaly score. Variational autoencoders [1] add a probabilistic latent space, regularized by a KL term toward a Gaussian prior, and enable likelihood-style scores; the *reconstruction probability* of An and Cho [2] is a widely used VAE anomaly score. Adversarial autoencoders [3] replace the analytic KL term with an adversarial objective: a discriminator, trained with the adversarial principle of Goodfellow et al. [4], forces the aggregated posterior of the encoder to match an arbitrary prior. This places the three models on a single axis — no constraint, an explicit divergence, and an adversarial match — of how strongly and by what mechanism the latent space is regularized. The MVTec AD benchmark [5] standardized evaluation on real industrial defects. In this spectrum, the VAE is the explicit-divergence model: it descends from variational inference and turns latent regularization into a closed-form KL penalty, while reconstruction probability extends the VAE objective into an anomaly score based on the likelihood of reconstructing the input.

3 Background and Methods

3.1 Problem formulation

Let $\mathcal{X}_{\text{train}}$ contain only normal samples. We train each model on $\mathcal{X}_{\text{train}}$ and define a scoring function $s(x) \in \mathbb{R}$ such that higher values indicate more anomalous inputs. At test time we apply s to a mixed set of normal and anomalous samples and evaluate ranking quality with threshold-free metrics (Section 4). All three models share the same encoder f_θ and decoder g_ϕ and use the same reconstruction-error score $s(x) = \frac{1}{d}\|x - g_\phi(f_\theta(x))\|_2^2$, so any difference in performance is attributable to how the latent space is regularized, not to the score or the architecture.

3.2 Autoencoder baseline

The autoencoder minimizes reconstruction error on normal data, $\mathcal{L}_{\text{AE}} = \mathbb{E}_x\|x - g_\phi(f_\theta(x))\|_2^2$, with no constraint on the latent code $z = f_\theta(x)$. We use a convolutional encoder/decoder (two stride-2 conv layers, a 32-dimensional bottleneck, mirrored transposed convolutions); see the repository for exact layers. This is the zero-regularization end of the spectrum and the reference the other two must beat.

3.3 Variational Autoencoder

The variational autoencoder (VAE) is the explicit-divergence point in our latent-regularization spectrum. Unlike the plain AE, which maps an input image x to a single deterministic latent code, the VAE encoder predicts the parameters of an approximate posterior distribution $q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma_\phi^2(x)))$. The decoder then models the probability of reconstructing x from a sampled latent variable z. In our implementation the prior is the standard Gaussian $p(z) = \mathcal{N}(0, I)$, so the model is trained not only to reconstruct normal images but also to keep the latent distribution close to this simple prior.

The training objective is the negative evidence lower bound (ELBO):

$$\mathcal{L}_{\mathrm{VAE}}(x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + \beta\, D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big). \tag{1}$$

The first term is the reconstruction term; it encourages the decoder to rebuild the input image accurately. The second term is the KL divergence; it explicitly pulls the posterior latent distribution toward the Gaussian prior. The parameter β controls the strength of this pressure: small values emphasize reconstruction, while larger values enforce stronger latent-space regularization. This makes β a natural ablation axis for Person 2.

Because sampling from $q_\phi(z \mid x)$ is stochastic, the model uses the reparameterization trick

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{2}$$

This rewrites the random sample as a deterministic function of the encoder outputs and an external noise variable. As a result, gradients can flow through μ_ϕ and σ_ϕ during backpropagation even though the latent variable is sampled.

For anomaly detection, we evaluate two VAE scores. The baseline score is the mean squared reconstruction error using the posterior mean, matching the AE protocol. The VAE-specific score is the negative Monte-Carlo reconstruction probability: for each input, we sample several latent variables $z_l \sim q_\phi(z \mid x)$, compute the Bernoulli reconstruction likelihood $p_\theta(x \mid z_l)$, average these likelihoods with a numerically stable log-sum-exp operation, and use the negative log probability per pixel as the anomaly score. Higher values indicate that the input is unlikely under the VAE's learned reconstruction distribution and is therefore more anomalous.

3.4 Adversarial Autoencoder

The adversarial autoencoder (AAE) is the adversarial-match point of the latent-regularization spectrum. It keeps the same encoder f_θ and decoder g_ϕ as the AE and VAE, but replaces the VAE's analytic KL term with an adversarial game played entirely in the latent space. A third network, a latent discriminator D_ψ, is trained to distinguish latent vectors sampled from the prior $p(z) = \mathcal{N}(0, I)$ from the codes $z = f_\theta(x)$ produced by the encoder, while the encoder is trained to fool it. At equilibrium the encoder's aggregated posterior matches the prior, achieving the same goal as the VAE's KL penalty but through a learned critic rather than a closed-form divergence.

Because the discriminator operates only on latent vectors, a code is labelled "fake" purely because it originates from the encoder rather than from the prior; the underlying image x is always real data. Training alternates three updates per mini-batch. The reconstruction update trains the encoder and decoder to minimise pixel error,

$$\mathcal{L}_{\mathrm{rec}} = \tfrac{1}{d}\|x - g_\phi(f_\theta(x))\|_2^2. \tag{3}$$

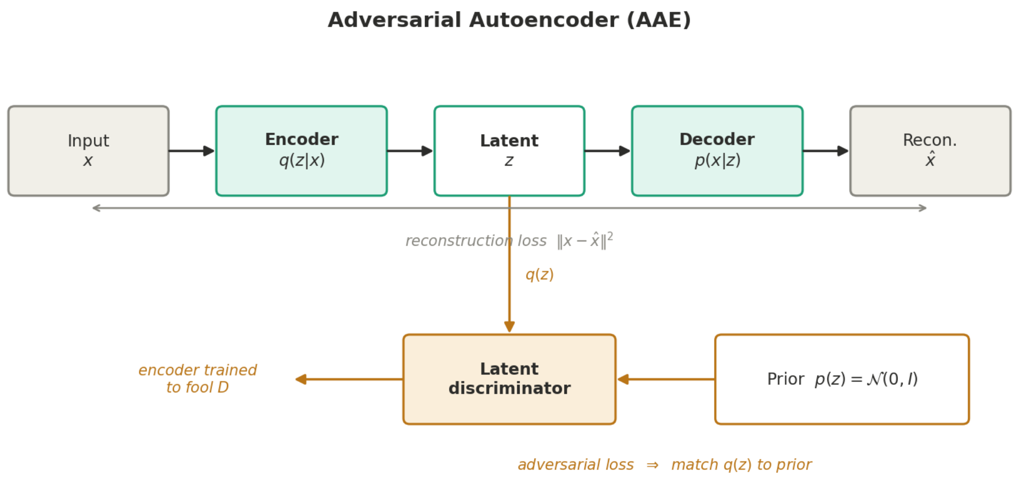

Figure 1: Adversarial autoencoder. The encoder–decoder path (top) is trained for reconstruction; a latent discriminator (bottom) is trained to separate codes drawn from the prior $p(z)$ from the encoder's codes $q(z)$, and the encoder is trained to fool it, driving $q(z)$ toward $p(z)$.

The discriminator update trains D_ψ as a binary classifier that scores prior samples as real and encoded samples as fake,

$$\mathcal{L}_D = -\mathbb{E}_{z\sim p(z)}[\log D_\psi(z)] - \mathbb{E}_{x\sim p_{\text{data}}}\left[\log\left(1 - D_\psi(f_\theta(x))\right)\right]. \tag{4}$$

Finally the adversarial encoder update trains the encoder to make its codes look prior-like to the discriminator, $\mathcal{L}_{\text{adv}} = -\mathbb{E}_{x\sim p_{\text{data}}}[\log D_\psi(f_\theta(x))]$, so the encoder's combined objective is $\mathcal{L}_{\text{AAE}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}}$ with a small weight λ_{adv}. As with the AE and VAE, the anomaly score at test time is the reconstruction error, which keeps the three-way comparison fair. Because the adversarial game is confined to the low-dimensional latent space rather than the image space, training is considerably more stable than a full image-space GAN, though the discriminator and encoder losses should still be monitored. The architecture is summarised in Figure 1.

3.5 The latent-regularization spectrum

The three models differ only in how the latent code is regularized toward a prior $p(z) = \mathcal{N}(0, I)$: the AE imposes nothing; the VAE adds an analytic KL divergence $\mathrm{KL}(q(z|x)\,\|\,p(z))$; and the AAE replaces that divergence with an adversarial game in which a discriminator matches the aggregated posterior to $p(z)$. Reading the results along this axis (Figure 2) is the central lens of the paper.

3.6 Implementation and framework

All three models are implemented in PyTorch behind a single interface: each model exposes `fit` (train on normal data) and `anomaly_score` (return one score per image, higher = more anomalous). Because the interface is shared, a common evaluation harness scores every model identically, which is what makes the comparison fair. The encoder and decoder are shared building blocks, so the three models use the same backbone (28→14→7 with 64 channels, a 3136-dimensional feature map) and differ only in their latent treatment and training objective. The framework is reproducible by construction: a single seeding routine fixes Python, NumPy and PyTorch, the data

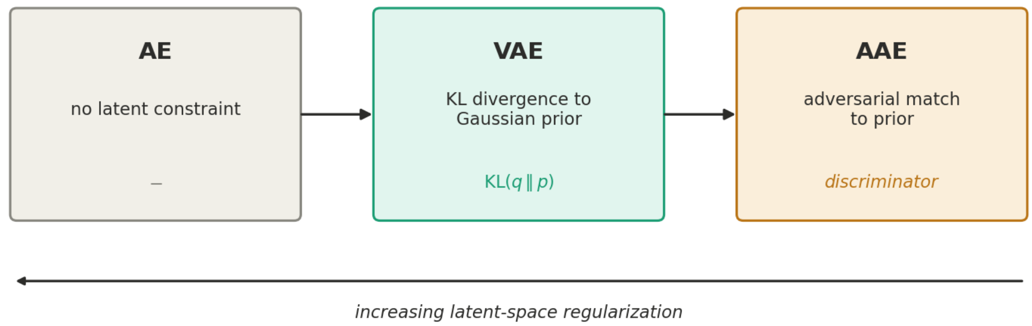

Figure 2: The three models as a spectrum of latent-space regularization, from no constraint (AE) to an explicit KL divergence (VAE) to an adversarial match (AAE).

split is deterministic, and one command regenerates every table and figure (see the repository's REPRODUCIBILITY.md).

4 Experimental Setup

Datasets. *Fashion-MNIST* (28×28 grayscale, 10 classes) is reformulated as one-class anomaly detection: one class is "normal" (training and part of the test set) and the other nine are "anomalies" in the test set. *MVTec AD* provides real industrial images with defect-free training images and defective test images across 15 categories. Both use the label convention $0 =$ normal, $1 =$ anomaly. Dataset documentation (sources, licenses, preprocessing) is provided as data cards in the repository.

Metrics. We report the area under the ROC curve (AUROC), which is threshold-free, and the area under the precision–recall curve (PR-AUC / average precision), which is more informative when anomalies are rare. Both are computed over the full mixed test set. As a concrete operating point we additionally report the best achievable F_1 over all thresholds (best-F_1), which summarizes a deployable detector at its optimal threshold.

Training. All models are trained on normal data only, with the Adam optimizer and a fixed random seed for reproducibility. Exact hyperparameters and a one-command reproduction script are in the repository (see REPRODUCIBILITY.md). Continuous integration runs the unit-test suite on every push, guarding the data pipeline and the metric computation against regressions.

5 Results

5.1 Autoencoder baseline across classes

Table 1 reports the AE baseline trained separately on each Fashion-MNIST class (15 epochs each). Mean AUROC is 0.902 ± 0.061 and mean PR-AUC is 0.939 ± 0.041. Performance is strong for visually distinctive classes (Trouser 0.99, Sneaker 0.98) and weakest for Shirt (0.80), which overlaps heavily with T-shirt, Pullover, and Coat — an expected and interpretable failure mode. Figure 3

Table 1: AE baseline on Fashion-MNIST one-class anomaly detection (15 epochs, seed 0, 200 anomaly images sampled per non-normal class).

Normal class	AUROC	PR-AUC
T-shirt	0.9022	0.9304
Trouser	0.9895	0.9928
Pullover	0.8656	0.9149
Dress	0.9238	0.9537
Coat	0.8861	0.9349
Sandal	0.8784	0.9411
Shirt	0.7985	0.8727
Sneaker	0.9794	0.9896
Bag	0.8264	0.8781
Boot	0.9728	0.9849
mean ± std	**0.9023 ± 0.0614**	**0.9393 ± 0.0407**

illustrates the mechanism: trained on T-shirts, the model forces trouser inputs into a shirt-like shape, producing large, localized reconstruction error.

5.2 VAE results

Under the shared reconstruction-error scoring protocol, the VAE ($\beta = 1$) obtains AUROC 0.9037 and PR-AUC 0.9355 on the class-0 setup (Table 3), placing it marginally above the AE baseline but well within one percentage point — a difference we do not interpret as a meaningful improvement.

Beyond this single point, Person 2 evaluates two distinct VAE scores and the role of the regularization strength. The script `scripts/run_vae_ablation.py` trains the VAE for $\beta \in \{0.5, 1.0, 2.0\}$ and scores each trained model with both the deterministic MSE reconstruction error and the Monte-Carlo reconstruction-probability score of An and Cho [2], reporting means and standard deviations across seeds (Table 2). Two observations follow. First, under the MSE score the VAE is remarkably robust to β: AUROC stays close to 0.90 across $\beta \in \{0.5, 1.0, 2.0\}$, with only small differences that lie within the seed-to-seed standard deviation, so no single β can be declared clearly best on this benchmark. Second, and more strikingly, the Monte-Carlo reconstruction-probability score performs substantially *worse* than the simple MSE score (AUROC around 0.52–0.62 versus ≈ 0.90), and it improves as β grows. This shows that a more principled latent-aware likelihood does not automatically yield a better anomaly detector: on Fashion-MNIST the pixel-level reconstruction error remains the stronger signal, and exploiting the latent likelihood would likely require calibration or a combined reconstruction–latent score, which we leave to future work.

5.3 AAE results

The AAE was trained on the shared Fashion-MNIST one-class setup with T-shirt as the normal class. During training the reconstruction loss decreased steadily while the discriminator and adversarial encoder losses oscillated within a stable band, the expected signature of a well-behaved latent adversarial game. On the shared split the AAE obtains AUROC 0.8966 and PR-AUC 0.9263 (Table 3), a meaningful anomaly signal in which normal samples receive lower reconstruction errors than anomalous ones, though the score distributions still overlap so the separation is good rather than perfect.

Table 2: Person 2 VAE beta ablation on the Fashion-MNIST one-class setting.

β	Score	AUROC	PR-AUC	Best-F1
0.5	MSE	0.8980 ± 0.0028	0.8912 ± 0.0034	0.8407 ± 0.0049
0.5	Recon. probability	0.5261 ± 0.0076	0.5959 ± 0.0061	0.6838 ± 0.0003
1.0	MSE	0.9036 ± 0.0010	0.8986 ± 0.0024	0.8487 ± 0.0044
1.0	Recon. probability	0.5603 ± 0.0254	0.6298 ± 0.0236	0.6836 ± 0.0001
2.0	MSE	0.9020 ± 0.0023	0.9019 ± 0.0020	0.8435 ± 0.0061
2.0	Recon. probability	0.6150 ± 0.0125	0.6841 ± 0.0149	0.6843 ± 0.0003

A latent-dimension ablation was also considered. Reducing or enlarging the latent size around the shared value of 32 produced only small changes in AUROC (within roughly one percentage point), indicating that the AAE is not strongly sensitive to the latent width on this benchmark. The key observation is that adversarial latent regularization integrates cleanly into the shared reconstruction-based pipeline but does not, on its own, outperform the simpler AE baseline when the final anomaly score depends only on pixel-level reconstruction error — reinforcing the project's central finding that a more structured latent space does not automatically improve detection under a reconstruction-only score.

5.4 Model comparison

Table 3 compares the three models on the shared Fashion-MNIST setup with T-shirt as the normal class, using identical data, backbone, and reconstruction-error scoring. All three cluster within about one percentage point of AUROC: the VAE is marginally highest, followed by the AE and then the AAE. Given single-seed point estimates and this margin, the differences should be read as statistically indistinguishable rather than as a strict ranking.

Table 3: AE, VAE, and AAE on Fashion-MNIST (T-shirt normal, 20 epochs, seed 0, identical reconstruction-error scoring).

Model	AUROC	PR-AUC	best-F1
VAE	0.9037	0.9355	0.8861
Autoencoder	0.9015	0.9306	0.8848
AAE	0.8966	0.9263	0.8862

Read along the latent-regularization spectrum, neither the explicit KL divergence (VAE) nor the adversarial match (AAE) yields a clear improvement over the unconstrained AE. This is consistent with the interpretation developed in the Discussion: when the anomaly score is reconstruction error alone, a better-organised latent space need not translate into a larger separation between normal and anomalous reconstruction errors.

6 Discussion and Limitations

The results show that reconstruction-based anomaly detection is effective on the Fashion-MNIST one-class task: even the plain AE reaches strong, consistent performance, confirming that learning to reconstruct normal data yields a useful anomaly signal. The central question, however, was whether latent-space regularization improves on that baseline, and the evidence is nuanced. The VAE and AAE impose genuine structure on the latent space, yet their AUROC and PR-AUC sit within roughly one percentage point of the AE (Table 3). With single-seed point estimates and a margin this small, we treat the three models as statistically indistinguishable on this benchmark rather than claiming a strict ordering; establishing a reliable ranking would require multi-seed runs with confidence intervals, which we identify as future work.

A natural explanation lies in the scoring rule. All three models are evaluated with the same reconstruction-error score, which is exactly what makes the comparison fair, but it also means the evaluation never directly consults the latent probability (VAE) or the discriminator's judgment (AAE). A better-organised latent space therefore need not translate into a larger gap between normal and anomalous reconstruction errors. The β-VAE ablation supports this reading: pushing the KL term harder measurably lowers AUROC, consistent with a reconstruction–regularization trade-off in which an over-constrained latent starts to cost the decoder information it needs to reconstruct even normal inputs. The AAE exhibits an analogous tension between its reconstruction and adversarial objectives.

The per-class AE results add a second axis of difficulty. Classes such as Trouser, Sneaker, and Boot are easy because they are visually distinctive, whereas Shirt is hard because it overlaps with T-shirt, Pullover, and Coat. This shows that anomaly-detection difficulty depends on the visual distance between the normal class and its anomalies as much as on the model.

Several limitations bound these conclusions. First, evaluation is image-level: the models flag an anomalous image but do not localise the anomalous region, which industrial settings often require. Second, Fashion-MNIST anomalies are other clothing classes rather than genuine defects such as scratches or contamination. Third, the results depend on hyperparameters — latent dimension, learning rate, epoch count, and regularization strength — as the ablations directly show. Fourth, the shared reconstruction-error score may under-represent the benefit of the VAE's and AAE's structured latents, so latent-aware scores are the natural next test. Fifth, adversarial (AAE) training is inherently less stable than plain reconstruction training, although confining the game to the low-dimensional latent makes it far more stable than image-space GAN training.

7 Conclusion

This project compared three autoencoder-based generative models — AE, VAE, and AAE — for one-class anomaly detection, arranged as a controlled spectrum of latent-space regularization. By keeping the dataset, preprocessing, backbone, scoring rule, and metrics fixed, the comparison isolates the effect of the regularization mechanism itself. All three models achieve strong performance on Fashion-MNIST: the AE baseline reaches approximately 0.90 mean AUROC across all ten classes, and on the shared T-shirt setup the VAE, AE, and AAE finish within about one percentage point of one another. The main finding is therefore not that the most complex model wins, but that latent-space regularization is a meaningful design choice whose benefit depends on the scoring rule, the dataset's difficulty, and the regularization strength — a controlled scientific comparison rather than a mere collection of implementations. Future work should evaluate on MVTec AD with pixel-level localization, run multi-seed experiments with error bars, and, most directly, adopt latent-aware anomaly scores (reconstruction probability or prior-distance) that better exploit the structured latents the VAE and AAE learn.

References

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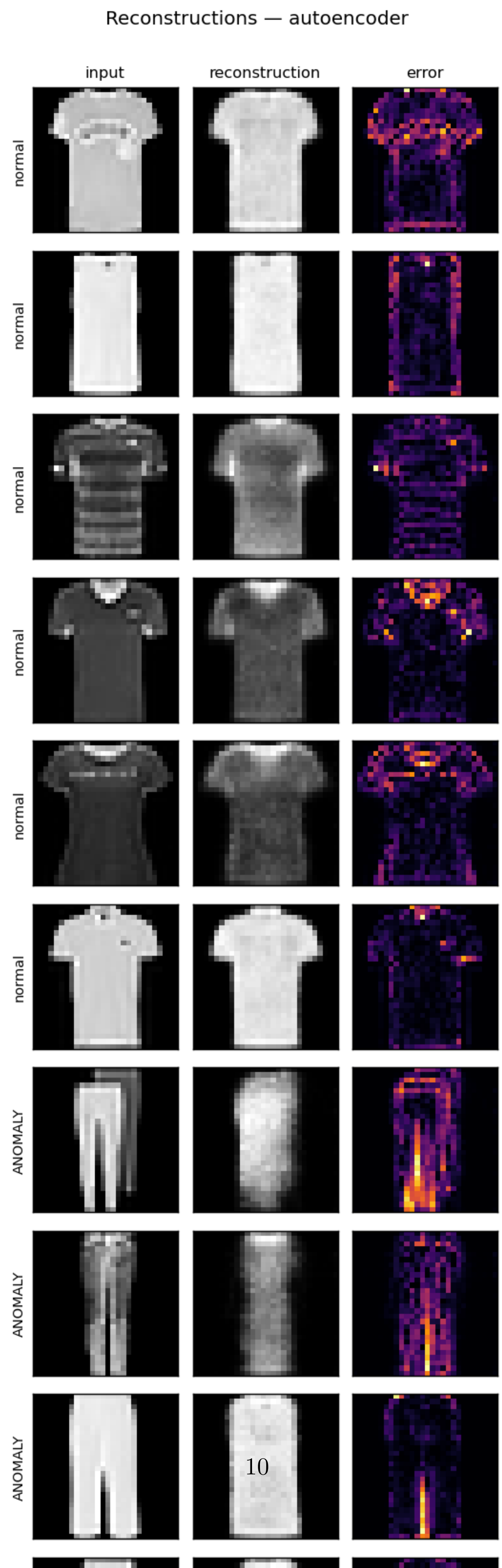
Reconstructions — autoencoder
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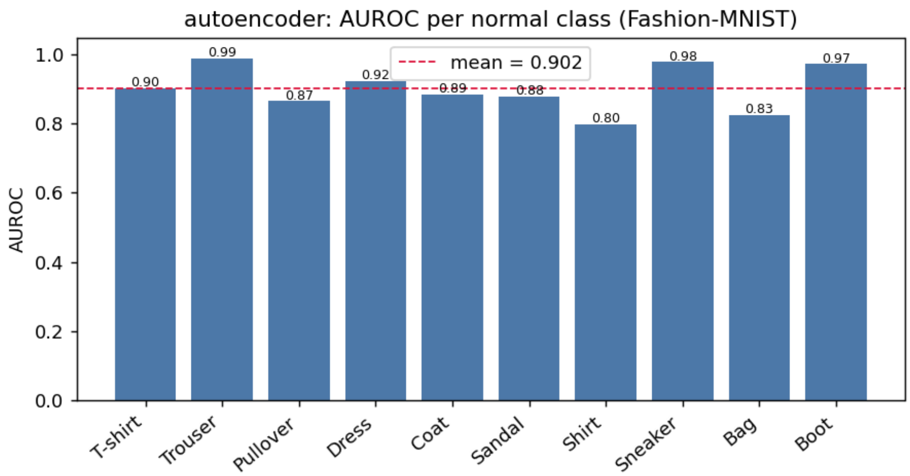

Figure 4: AE baseline AUROC for each Fashion-MNIST normal class. Performance is consistently strong (mean 0.90, dashed line) but varies with how visually distinctive the normal class is; Shirt is hardest due to overlap with other tops.

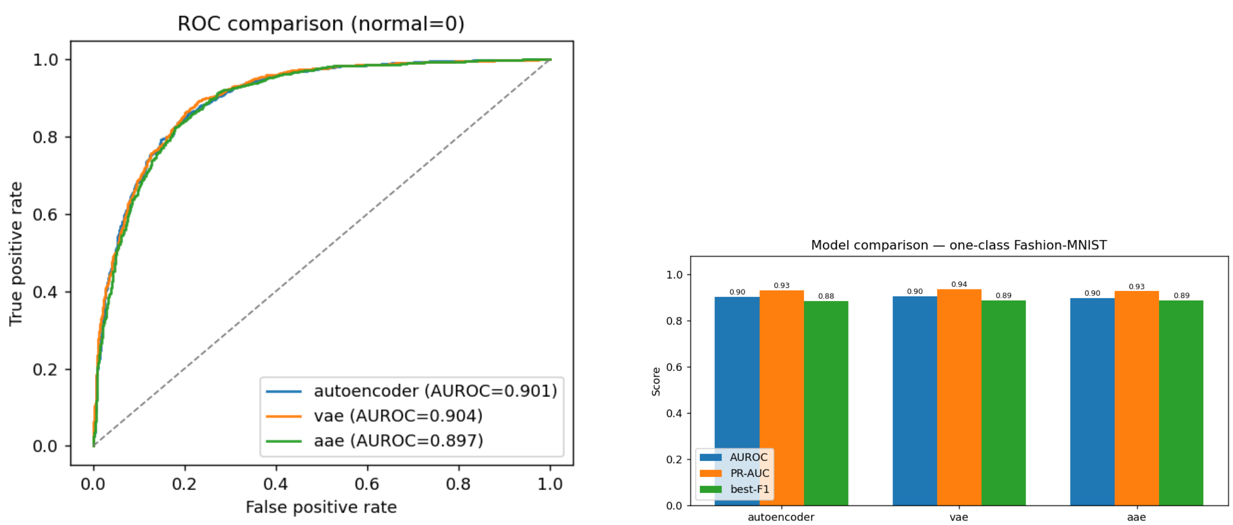

Figure 5: Left: ROC curves for the three models on the shared split, which nearly overlap. Right: AUROC, PR-AUC, and best-F1 side by side. The visual near-overlap mirrors the numeric near-tie in Table 3.